TwinEarth AI — The Living Digital Twin of Planet Earth

Date: May 1, 2026 — Evening Drop
Category: Earth Intelligence / Climate Tech / Geospatial AI / Defense & Government
TAM: $312B (Earth Observation $89B, Climate Analytics $67B, Geospatial Intelligence $78B, Digital Twin Infrastructure $78B)

The Problem: We're Flying Blind on Our Own Planet

Earth is the most important system humanity depends on — and we have no real-time, unified model of how it works.

We have: - **18,000+ satellites** collecting exabytes of Earth data - **Millions of IoT sensors** monitoring weather, water, soil, air - **Countless research institutions** building isolated models - **Governments and enterprises** making trillion-dollar decisions

Yet there's **no single, coherent, real-time digital twin of Earth**.

The Fragmentation Catastrophe

Problem	Impact	Annual Cost
Siloed satellite data	Same imagery processed 50+ times by different orgs	$34B in duplicated effort
No unified Earth model	Climate models don't talk to agricultural models	$89B in misallocated resources
Reactive disaster response	We detect hurricanes but can't model cascading effects	$178B in preventable damage
Commodity blind spots	Can't see global food/resource flows in real-time	$56B in supply chain shocks
Insurance mispricing	Climate risk models are years out of date	$112B in misallocated capital

The real cost: We're making civilization-scale decisions — where to build cities, how to price risk, when to plant crops, how to allocate water — using fragmented, stale, incompatible data.

Why Now?

1. **Satellite constellations have matured** — Planet, Maxar, Spire, ICEYE provide daily global coverage
2. **Foundation models can fuse multimodal Earth data** — Google's weather AI, NVIDIA's Earth-2 prove the concept
3. **Compute is finally sufficient** — Training planetary-scale models is now economically viable
4. **Climate urgency demands it** — Insurers, governments, enterprises MUST understand Earth systems
5. **Regulatory tailwinds** — EU's Destination Earth, US climate disclosure rules mandate this intelligence

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The Solution: TwinEarth AI

TwinEarth is the operating system for planetary intelligence — a living, breathing digital twin of Earth that updates continuously and answers any question about our planet.

We're not building another satellite imagery company. We're building **the unified intelligence layer that makes all Earth data useful**.

The Architecture

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                          TWINEARTH AI PLATFORM



                      PLANETARY FOUNDATION MODEL
            Unified multimodal understanding of Earth systems
       (Climate • Hydrology • Agriculture • Infrastructure • Ecology)




   OPTICAL       RADAR         LIDAR         THERMAL       IOT
  Satellites    Satellites    Aircraft      Satellites    Sensors
  (Planet,      (ICEYE,       (Drone        (Landsat,     (Ground,
   Maxar)        Capella)      Fleets)       GOES)         Buoys)



                          INTELLIGENCE MODULES

  CLIMATE      AGRICULTURE   WATER        INFRA-        BIODIVERSITY
  ENGINE       ENGINE        ENGINE       STRUCTURE     ENGINE

 • Weather     • Crop        • Aquifer    • Building    • Species
   Forecast      Health        Levels       Detection     Tracking
 • Climate     • Yield       • Flood      • Road        • Habitat
   Scenarios     Predict       Risk         Condition     Health
 • Extreme     • Soil        • Drought    • Power       • Deforestation
   Events        Carbon        Monitor      Grid          Alert
 • Carbon      • Pest        • Ocean      • Urban       • Migration
   Tracking      Detection     Health       Growth        Patterns
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Core Capabilities

1. Unified Earth Model

- Fuses data from 200+ satellite constellations, 10M+ ground sensors
- Updates every 15 minutes for dynamic layers, daily for static
- 10-meter resolution globally, 30cm in urban areas
- Historical depth back to 1984 (Landsat archive integration)

2. Natural Language Earth Queries Ask questions in plain English: - *"Show me all areas in Brazil at risk of deforestation in the next 90 days"* - *"What's the current wheat harvest status in Ukraine compared to*

5-year average?" - *"Identify all coastal infrastructure at risk from sea level rise by 2035"* - *"Track this cargo ship's likely path and predict arrival"*

3. Scenario Simulation

- Run "what if" scenarios on Earth systems
- *"What happens to California water supply if snowpack drops 30%?"*
- *"Model cascading effects of a Category 5 hurricane hitting Miami"*
- *"Simulate crop yields under 2°C warming scenario"*

4. Anomaly Detection at Planetary Scale

- Automatic detection of changes: construction, deforestation, military activity
- Early warning for disasters: fires, floods, droughts, disease outbreaks
- Supply chain visibility: track global commodity flows in real-time

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Go-To-Market Strategy

Phase 1: Anchor Customers (Months 1-12)

Target: Reinsurance Giants - Munich Re, Swiss Re, Lloyd's spend $2B+/year on climate risk modeling - Their models are fragmented, stale, and siloed - **Value prop:** Real-time global risk assessment, 10x faster scenario modeling - **Entry point:** $5M annual platform license + usage - **Expansion:** Embed in underwriting workflows

Target: Commodity Trading Houses - Glencore, Cargill, Trafigura need real-time crop/resource intelligence - Current data is delayed, incomplete, expensive - **Value prop:** See global supply/demand shifts days before markets - **Entry point:** $3M annual + alpha-generating signals - **Expansion:** Operational planning, logistics optimization

Phase 2: Government & Defense (Months 6-18)

Target: Defense & Intelligence - NGA, NRO, Five Eyes agencies need unified GEOINT - **Value prop:** Persistent global monitoring, automatic change detection - **Entry point:** SBIR/STTR contracts, pilot programs - **Expansion:** Enterprise deployment, custom classified instances

Target: Climate Agencies - NOAA, EPA, European Environment Agency - **Value prop:** Unified climate monitoring, policy impact modeling - **Entry point:** Research partnerships, public-sector pricing

Phase 3: Enterprise Scale (Months 12-24)

Vertical Modules: - AgriTwin — Precision agriculture for enterprise farms - **InsureTwin** — Parametric insurance pricing engine - **SupplyTwin** — Global supply chain visibility - **CarbonTwin** — Verified carbon credit monitoring

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Business Model

Revenue Streams

Stream	Description	Pricing	Year 3 Target
Platform License	Core TwinEarth access	$1-10M/year	$89M
API Calls	Query-based pricing	$0.01-$1.00/query	$34M

Stream	Description	Pricing	Year 3 Target
Custom Layers	Vertical-specific modules	$500K-$2M/year	$45M
Scenario Compute	Simulation runtime	$0.10-$10/sim-hour	$23M
Data Partnerships	Revenue share with providers	20% of data revenue	$12M
Government Contracts	Defense/Intel deployments	$5-50M/contract	$67M

Unit Economics

- **Gross Margin:** 78% (compute-heavy but highly leveraged)
- **CAC:** $450K (enterprise/government sales)
- **LTV:** $8.2M (5-year average contract value)
- **LTV/CAC:** 18.2x
- **Payback:** 11 months

Technical Architecture

The Planetary Foundation Model

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                    TWINEARTH FOUNDATION MODEL
                    (2.3 Trillion Parameters)


INPUT ENCODERS                    OUTPUT HEADS

• Optical imagery encoder         • Segmentation
• SAR/Radar encoder               • Object detection
• Time series encoder             • Change detection
• Spectral encoder                • Prediction/Forecast
• Text/metadata encoder           • Natural language
• Geospatial position encoder     • Scenario generation

CORE ARCHITECTURE

• Mixture-of-Experts backbone (efficient sparse activation)
• Spatial-temporal attention (captures Earth dynamics)
• Multi-resolution processing (10m to continental scale)
• Physics-informed constraints (Earth system priors)
• Continuous learning from new satellite passes
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Data Pipeline

Daily Data Volume: - 50 TB optical imagery - 12 TB SAR data - 8 TB thermal/spectral - 200 GB IoT sensor feeds - 50 GB weather station data - 100 GB shipping/AIS data

Processing Stack: - Kubernetes on GPU clusters (NVIDIA H100s) - Distributed training on 4,096+ GPUs - Edge inference for latency-sensitive queries - Multi-cloud deployment (AWS, GCP, Azure GOV)

Key Technical Differentiators

1. **Self-Supervised Pre-Training on Earth Archive**
   - Trained on 40+ years of Landsat, Sentinel, MODIS data
   - Learns Earth system dynamics without labels
   - Emergent understanding of seasons, weather, human activity
2. **Real-Time Fusion Engine**
   - Fuses data from multiple satellites within minutes
   - Handles cloud cover by combining optical + SAR
   - Temporal interpolation fills gaps
3. **Physics-Informed Architecture**
   - Conservation laws embedded in model
   - Physically plausible predictions
   - Uncertainty quantification built-in

Competitive Moat

Why We Win

Competitor	Limitation	Our Advantage
Planet Labs	Imagery only, no intelligence	We build the brain, they're one input
Descartes Labs	Narrow verticals, legacy ML	Foundation model approach, unified platform
Google Earth Engine	Research tool, not enterprise-ready	Purpose-built for commercial/government
Orbital Insight	Point solutions, no unified model	Holistic Earth understanding
NVIDIA Earth-2	Climate simulation only	Full-stack Earth intelligence
Destination Earth (EU)	Government project, years away	Commercial velocity, enterprise focus

Defensibility Layers

1. **Data Flywheel**
   - More customers → more queries → better understanding of what matters
   - Proprietary training signal from commercial usage
2. **Model Compound Learning**
   - Every satellite pass improves the model
   - Historical archive provides unique training advantage
   - Competitors can't catch up without the same history
3. **Network Effects**
   - Enterprises want the same Earth model as their partners
   - Standardization drives adoption
   - Switching costs increase over time
4. **Regulatory Capture**
   - Become the standard for climate disclosure
   - Embed in insurance regulatory frameworks
   - Government contracts create barriers

Financial Projections

Revenue Trajectory

Year	ARR	Customers	Key Milestone
2026	$8M	12	Platform launch, first reinsurers
2027	$45M	67	Defense contracts, AgriTwin launch
2028	$156M	234	Enterprise scale, global expansion
2029	$412M	890	Platform standard, API ecosystem
2030	$890M	2,400	Public markets trajectory

Funding Strategy

Round	Amount	Timing	Use of Funds
Seed	$8M	Q2 2026	Core team, initial model training
Series A	$45M	Q1 2027	Platform build, first customers
Series B	$120M	Q4 2027	Scale compute, defense certifications
Series C	$300M	Q3 2028	Global expansion, acquisitions

Path to $1B Valuation

- **Series A:** $45M at $180M post (4x on early traction)
- **Series B:** $120M at $600M post (defense + enterprise momentum)
- **Series C:** $300M at $1.5B+ post (category leadership)

Team Requirements

Founding Team (Ideal Profiles)

CEO — Geospatial Intelligence Veteran - Background: NGA, Planet Labs, or Maxar leadership - Skills: Government sales, enterprise GTM, vision - Key: Understands both defense and commercial markets

CTO — Foundation Model Expert - Background: Google AI, Meta FAIR, DeepMind - Skills: Large-scale model training, multimodal AI - Key: Has shipped planetary-scale ML systems

Chief Scientist — Earth System Expert - Background: NASA, NOAA, or climate research - Skills: Earth science, physical modeling - Key: Bridges AI and domain expertise

VP Engineering — Geospatial Infrastructure - Background: Planet, Google Earth, Esri - Skills: Satellite data pipelines, real-time systems - Key: Built production Earth observation systems

Key Early Hires

- ML Engineers (remote sensing, foundation models) — 8 FTEs
- Geospatial Engineers (data pipelines, fusion) — 6 FTEs
- Solutions Engineers (customer implementation) — 4 FTEs
- Defense BD (government contracts) — 2 FTEs
- Product Managers (vertical modules) — 3 FTEs

Risks & Mitigations

Risk	Severity	Mitigation
Compute costs explode	High	Optimize model efficiency, leverage edge inference, secure GPU allocations early
Satellite data access restricted	Medium	Multi-provider strategy, own ground stations, government partnerships
Defense contract timelines	Medium	Parallel commercial revenue, SBIR/STTR bridge funding
Google/NVIDIA competition	Medium	Speed to market, vertical specialization, customer lock-in
Talent scarcity	High	Remote-first, equity compensation, mission-driven culture
Data sovereignty regulations	Medium	Multi-region deployment, local partnerships, classified instances

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90-Day Sprint Plan

Days 1-30: Foundation

- ☐ Recruit CTO and Chief Scientist
- ☐ Secure $2M pre-seed from climate/defense VCs
- ☐ Begin Planetary Foundation Model architecture design
- ☐ Establish data partnerships (Planet, Spire, ICEYE)
- ☐ Identify 3 pilot reinsurance customers

Days 31-60: Build

- ☐ Start model training on historical satellite archive
- ☐ Build data ingestion pipeline (50TB/day capacity)
- ☐ Design natural language query interface
- ☐ Develop first vertical module (InsureTwin prototype)
- ☐ Begin government BD (schedule NGA, NOAA meetings)

Days 61-90: Validate

- ☐ Deploy alpha platform to 2 pilot customers
- ☐ Demonstrate real-time anomaly detection
- ☐ Secure first paid pilot ($500K)
- ☐ Submit SBIR Phase I proposals
- ☐ Prepare Seed deck with customer validation

Why This Wins

TwinEarth isn't just another satellite imagery company — it's the unified intelligence layer for understanding our planet.

The Inevitable Future

Earth observation is following the same trajectory as other data domains: - **Phase 1:** Collect data (done — thousands of satellites) - **Phase 2:** Process data (in progress — individual analytics) - **Phase 3:** Unified intelligence (TwinEarth — the opportunity)

Just as Snowflake unified enterprise data, **TwinEarth will unify Earth data**.

The Timing Is Perfect

1. **Foundation models work** — GPT, Claude, Gemini proved the approach
2. **Earth data is abundant** — Petabytes available, underutilized
3. **Climate urgency is real** — Trillions at stake, decisions must improve
4. **Government funding is flowing** — Destination Earth, US climate initiatives
5. **Enterprises are ready to pay** — Insurance, agriculture, commodities need this

The Prize

- **Immediate:** $312B TAM across verticals
- **Strategic:** Become the standard for Earth intelligence
- **Long-term:** Every decision about the physical world routes through TwinEarth

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The Ask

We're seeking: - $8M Seed to build the Planetary Foundation Model - Introductions to reinsurance, commodity trading, and defense buyers - Technical advisors with Earth observation and foundation model expertise

What you get: - Ground floor in the company that will own planetary intelligence - Category-defining opportunity with clear path to $10B+ outcome - Mission-driven investment: literally helping humanity understand Earth

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"We've mapped the genome, connected humanity, and built AI that thinks. Now it's time to truly understand the planet we live on."

TwinEarth AI — See the Whole Picture.

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